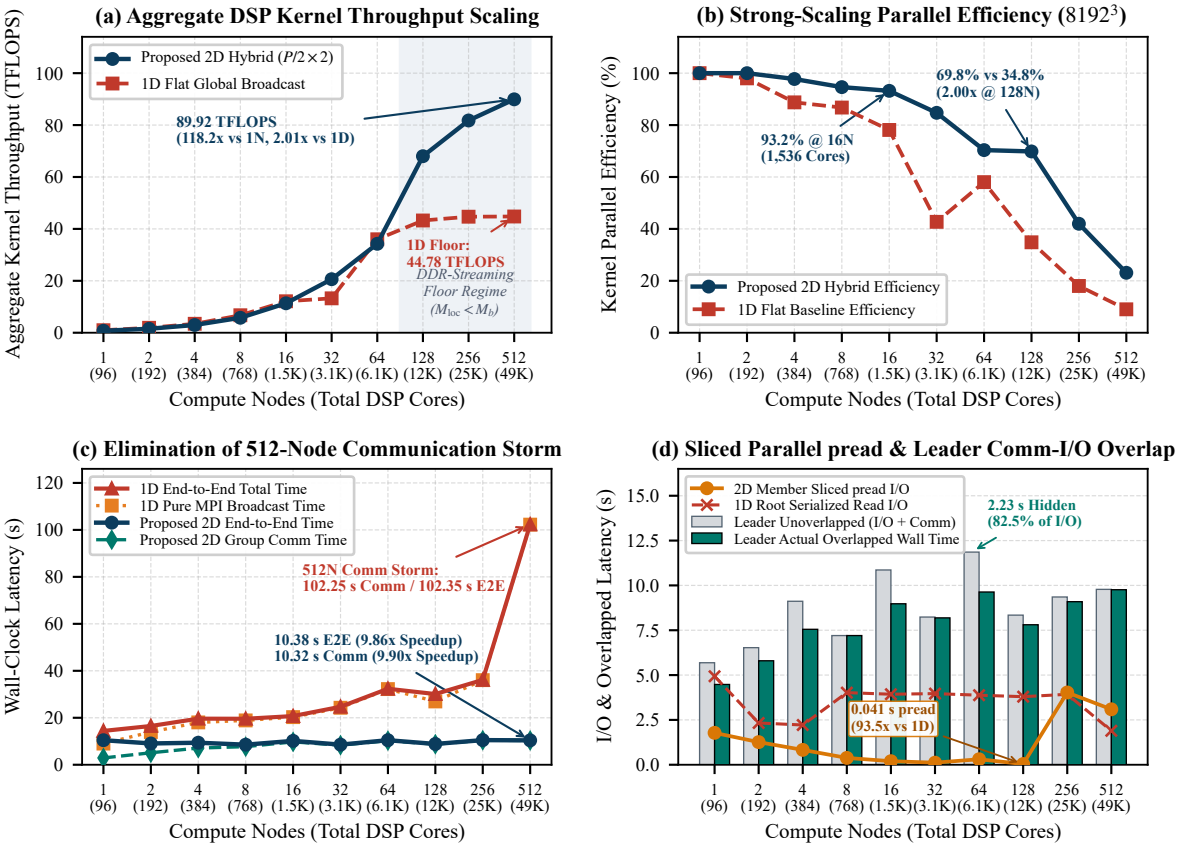

(a) Aggregate DSP Kernel Throughput Scaling
Proposed 2D Hybrid (P/2×2)
1D Flat Global Broadcast
89.92 TFLOPS
(118.2x vs 1N, 2.01x vs 1D)
1D Floor:
44.78 TFLOPS
DDR-Streaming
Floor Regime
(Mloc < Mb)
Aggregate Kernel Throughput (TFLOPS)
Compute Nodes (Total DSP Cores)
(b) Strong-Scaling Parallel Efficiency (8192³)
69.8% vs 34.8%
(2.00x @ 128N)
93.2% @ 16N
(1,536 Cores)
Proposed 2D Hybrid Efficiency
1D Flat Baseline Efficiency
Kernel Parallel Efficiency (%)
Compute Nodes (Total DSP Cores)
(c) Elimination of 512-Node Communication Storm
1D End-to-End Total Time
1D Pure MPI Broadcast Time
Proposed 2D End-to-End Time
Proposed 2D Group Comm Time
512N Comm Storm:
102.25 s Comm / 102.35 s E2E
10.38 s E2E (9.86x Speedup)
10.32 s Comm (9.90x Speedup)
Wall-Clock Latency (s)
Compute Nodes (Total DSP Cores)
(d) Sliced Parallel pread & Leader Comm-I/O Overlap
2D Member Sliced pread I/O
1D Root Serialized Read I/O
Leader Unoverlapped (I/O + Comm)
Leader Actual Overlapped Wall Time
2.23 s Hidden
(82.5% of I/O)
0.041 s pread
(93.5x vs 1D)
I/O & Overlapped Latency (s)
Compute Nodes (Total DSP Cores)